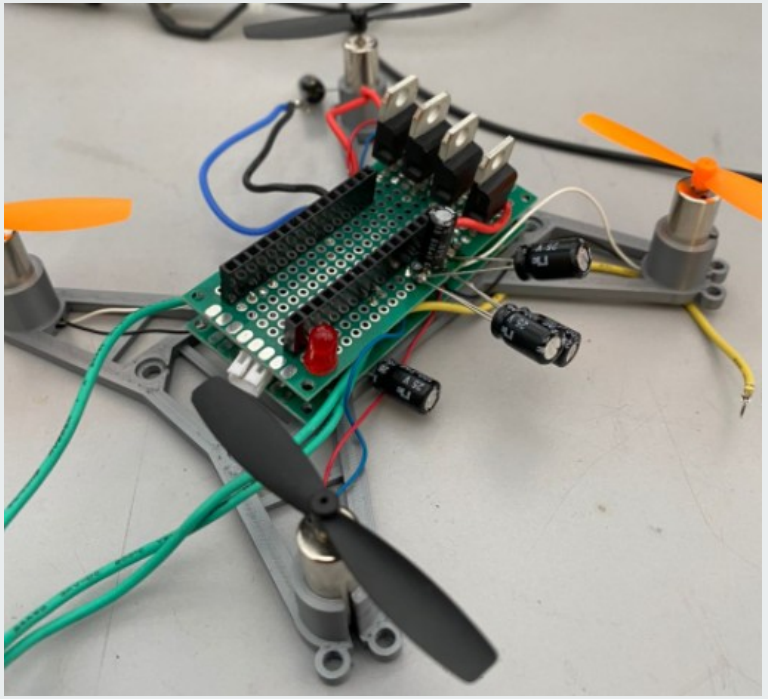
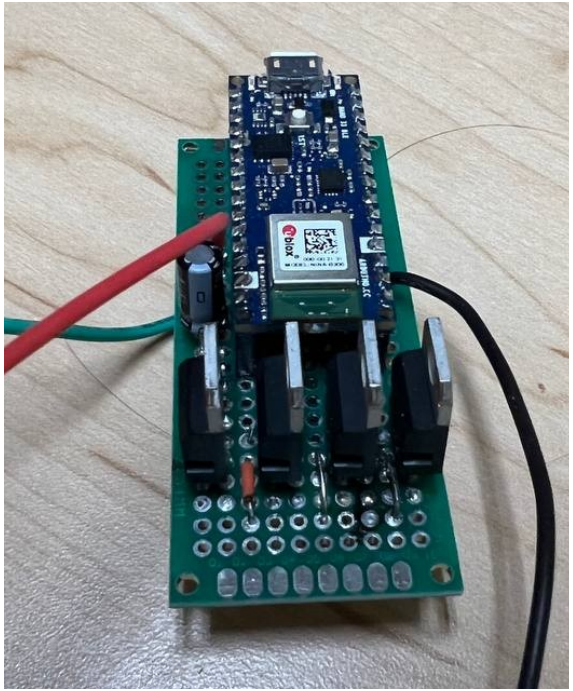
Nano Drone

Harkiret Singh



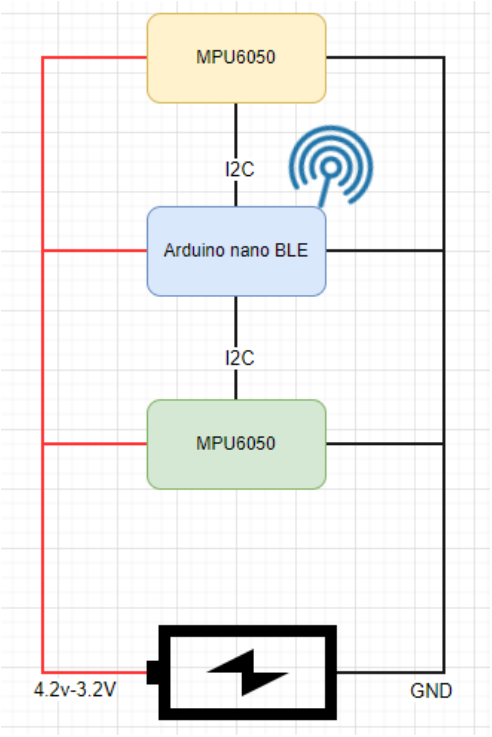
Main Features:

- Reverse battery protection system
- Wireless BLE connectivity
- Battery level checker
- Stabilizing motors using a PID controller
- GYRO implementation by MPU6050



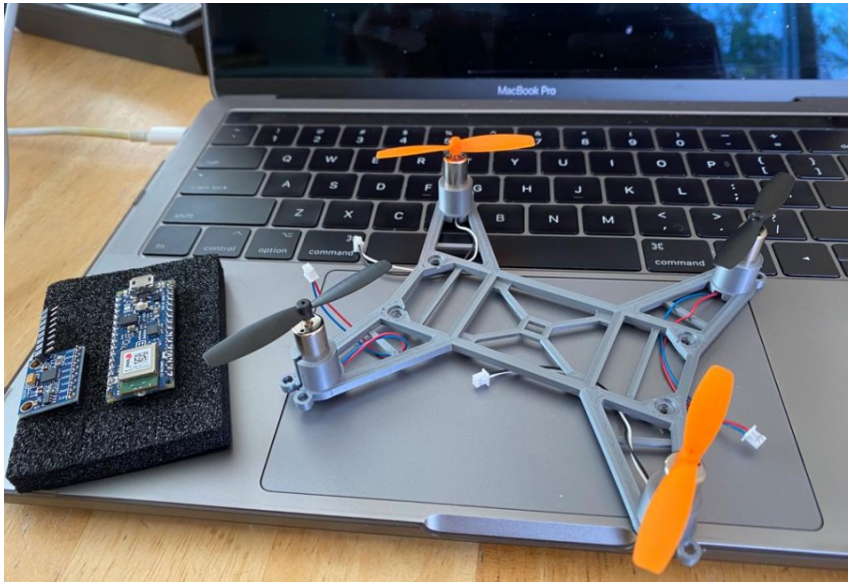
Main circuit

- Ardruino Nano 33 BLE
- MPU 6050 IMU
- VL53L0X Altitude sensor
- 4.2V 650MaH Lipo battery
- 8.5mm x 20mm motors DC Brushed motors and propellors
- P-mosfets and N-Mosfets
- Capacitors, resistors and diodes



Arduino nano BLE

- Has enough PWM output pins
- Has BLE connectivity module built in
- Easy to program
- Relatively small and light
- Future proof



BLE (Bluetooth Low Energy) Connectivity

- Custom 128-bit service & characteristics
- Characteristics hold the information

```
const char* deviceServiceUuid = "faa94135-6303-4274-a2ce-8fa28849ac9c";
const char* deviceServiceCharacteristicUuid = "a6671089-7187-4a45-adee-6020212f014d";

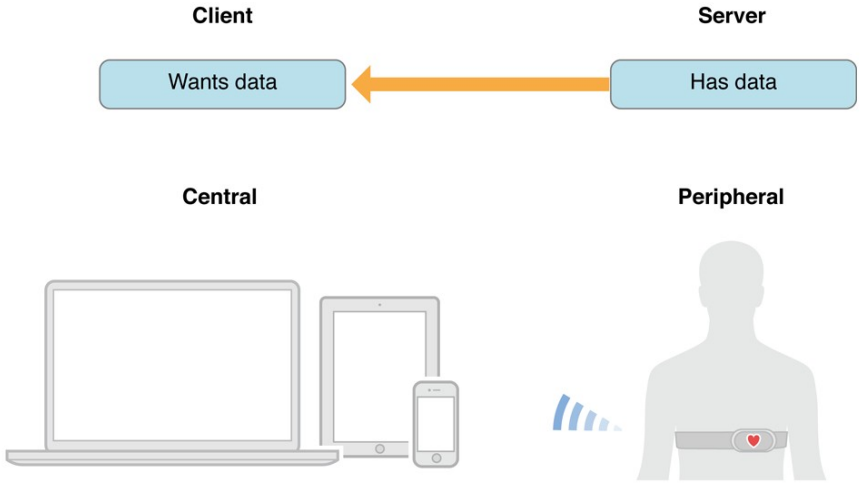
BLEService gestureService(deviceServiceUuid);
BLEByteCharacteristic gestureCharacteristic(deviceServiceCharacteristicUuid, BLERead | BLEWrite);

 BLE.setLocalName("Drone");
 BLE.setAdvertisedService(gestureService);
 gestureService.addCharacteristic(gestureCharacteristic);
 BLE.addService(gestureService);
 gestureCharacteristic.writeValue(-1);
 BLE.advertise();
```

BLE (Bluetooth Low Energy) Connectivity

- Connects to the custom app on our phone

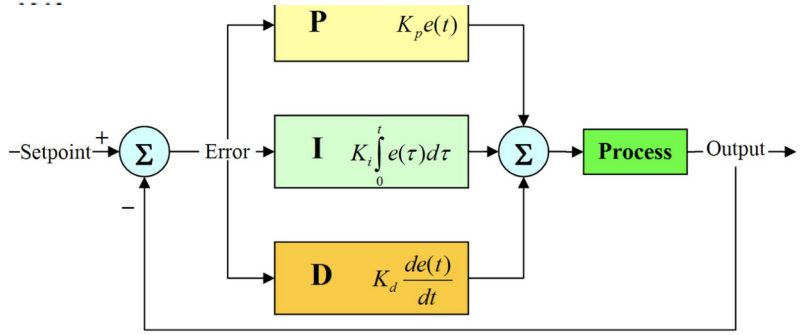
```
while (central.connected()) {
  if (gestureCharacteristic.written()) {
     gesture = gestureCharacteristic.value();
     writeGesture(gesture);
   }
}
```



PID controller

```
// Define Variables we'll be connecting to
double pitchSetpoint, pitchInput, pitchOutput;
double rollSetpoint, rollInput, rollOutput;

//Define the aggressive and conservative Tuning Parameters
double consKp = 1, consKi = 0.05, consKd = 0.25;
PID pitchPID(&rollInput, &rollOutput, &rollSetpoint, consKp, consKi, consKd, DIRECT);
PID rollPID(&pitchInput, &pitchOutput, &pitchSetpoint, consKp, consKi, consKd, DIRECT
```




```
  pitchPID.SetOutputLimits(-20, 20);
  rollPID.SetOutputLimits(-20, 20);

  targetSpeed[0] = -10;
  targetSpeed[1] = -10;
  targetSpeed[2] = -10;
  targetSpeed[3] = -10;
}

void changeTargetSpeed(int delta) {
  targetSpeed[0] += delta;
  targetSpeed[1] += delta;
  targetSpeed[2] += delta;
  targetSpeed[3] += delta;
}

void loop() {

  char bluetoothVal = 'a';

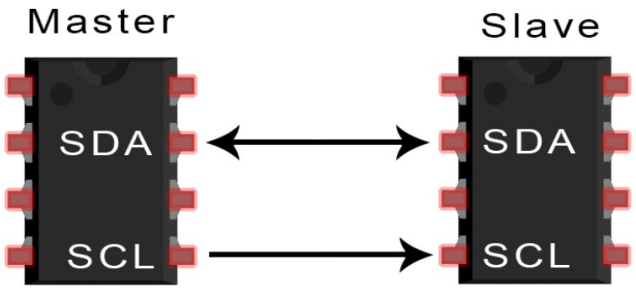
  if (Serial.available()) {
    bluetoothVal = Serial.read();
  }

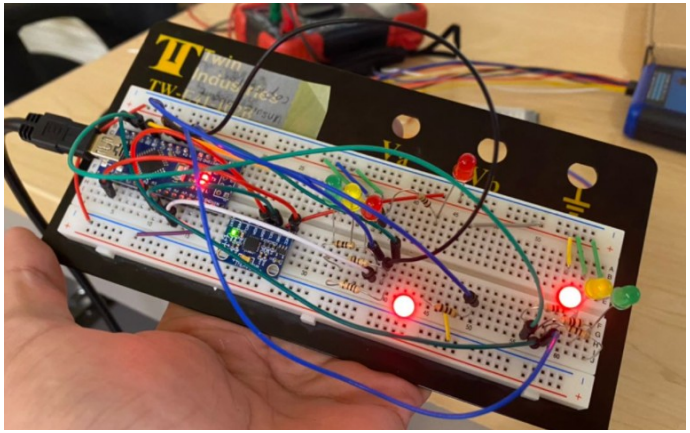
  if (bluetoothVal == 'u') {
    changeTargetSpeed(1);
  }

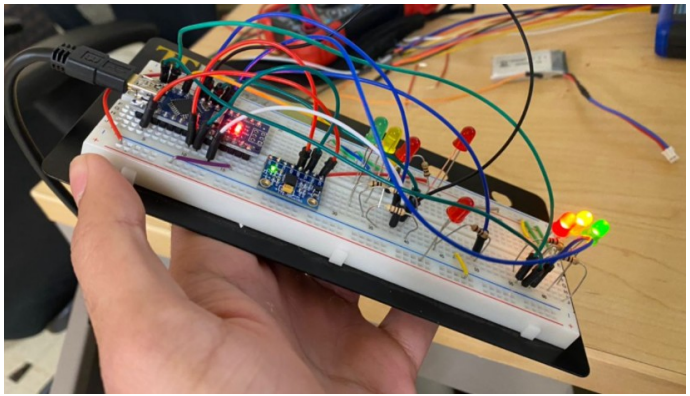
  if (bluetoothVal == 'd') {
    changeTargetSpeed(-1);
  }
```

LED replication of a GYRO

- MPU 6050 integration with Arduino Nano
- Communicates using I2C
- A protocol intended to allow multiple "peripheral" digital integrated circuits ("chips") to communicate with one or more "controller" chips



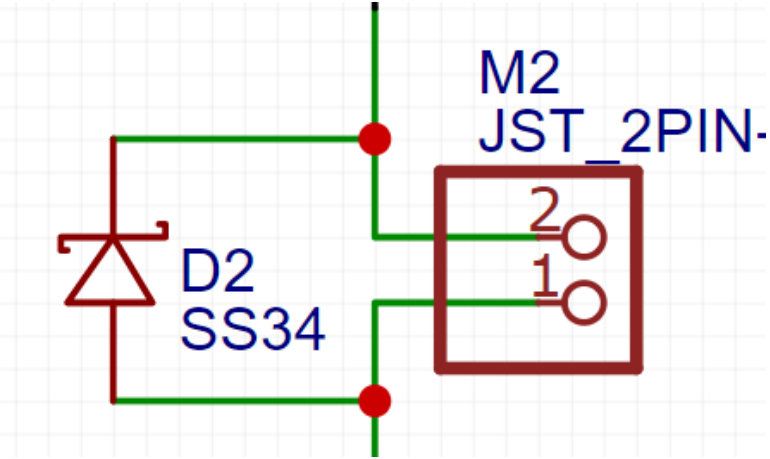


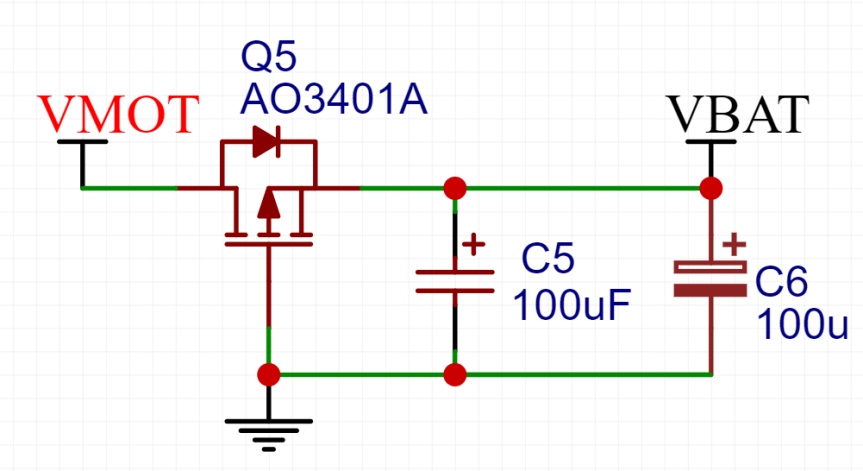


Gyroscope LED Control

- Representation of how the motors would counteract the movements
- Replicated using LED lights
- pitch/yaw for each erection
- LED’s get more bright as direction increases

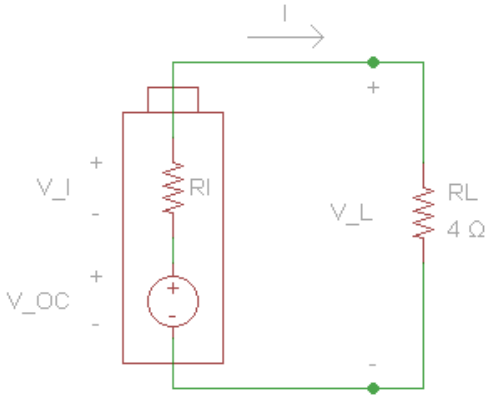
System protection

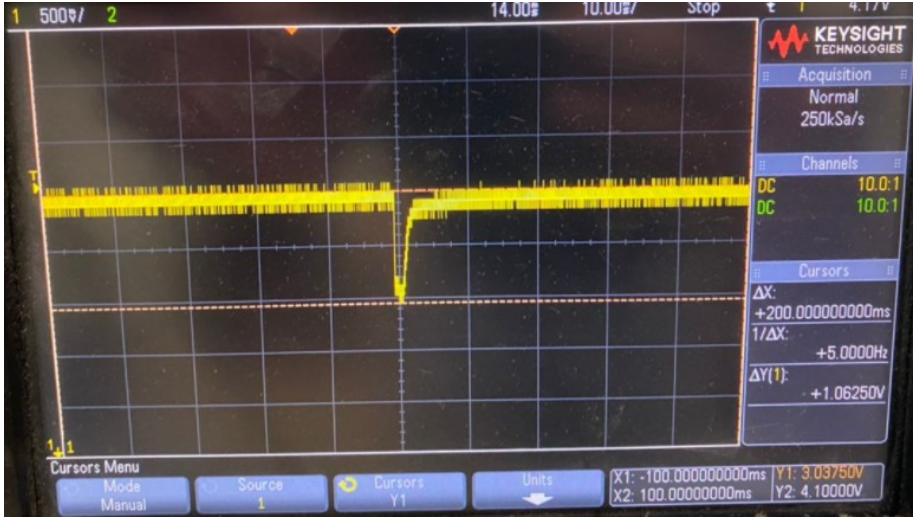




Problems encountered

- LIPO batteries dipped to 1.9V due to internal resistance
- Load current from each motor: 2.8Amps (11.2A)





BOM

Arduino Nano 33 BLE : $27.29

Connectors & ToF sensor: $35

Battery: $20.76

Motors : $27.32

Propellors: $16.38

Components from Mouser: $20.98

Components from Protosupplies: $23.40

Components from Anchor: $13.53